

# SIAI

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## Practical class #2

DHCP Servers in Linux

# IP Address Assignment for LANs

## **Three alternatives:**

- Manual configuration
- Via autoconfiguration (SLAAC/RFC 4862 in IPv6 or RFC3927 in IPv4).
- Via DHCP: Dynamic Host Configuration Protocol (most commonly used in IPv4)

# The road to DHCP

## The origins: **1980s – Manual configuration & BOOTP**

- Before DHCP, network administrators manually configured IP addresses for each device.
- In 1985, the **Bootstrap Protocol (BOOTP)** was standardized in [RFC 951]. BOOTP allowed diskless workstations to obtain an IP address and boot information from a central server.
- However, BOOTP had limitations: it required static configuration of MAC addresses on the server, making it inflexible for large or dynamic networks.

# The early years

## Early 1990s – Need for automation

- With the rapid expansion of TCP/IP networks and the internet, there was a need for a more flexible and automated way of assigning IP addresses.
- The **Dynamic Host Configuration Protocol (DHCP)** was designed as an extension of BOOTP.

## 1993 – First specification

- DHCP was formally defined in **RFC 1531** (October 1993).
- It introduced **dynamic leasing** of IP addresses, meaning devices could borrow an address for a limited time rather than requiring permanent assignments.

# Standardization

## **RFC 2131 (1997)**

- Replaced RFC 1531, refining the protocol.
- Defined the **DHCP Discover–Offer–Request–Acknowledge (DORA)** handshake still in use today.
- Provided support for dynamic allocation, automatic allocation, and manual allocation of addresses.

## **RFC 2132 (1997)**

- Defined the standard set of DHCP options (such as default gateway, DNS server, and domain name).

## **Widespread adoption (late 1990s–2000s)**

- Became the default method for IP configuration in enterprise, ISP, and home networks.
- Simplified network management drastically, especially with the rise of Wi-Fi and mobile devices

# DHCP, now

## **DHCPv6 (2003, RFC 3315)**

- Created for **IPv6**, since IPv6 addressing and autoconfiguration differ from IPv4.
- Provides more flexible options and can work alongside IPv6 Stateless Address Autoconfiguration (SLAAC)

## **Security considerations**

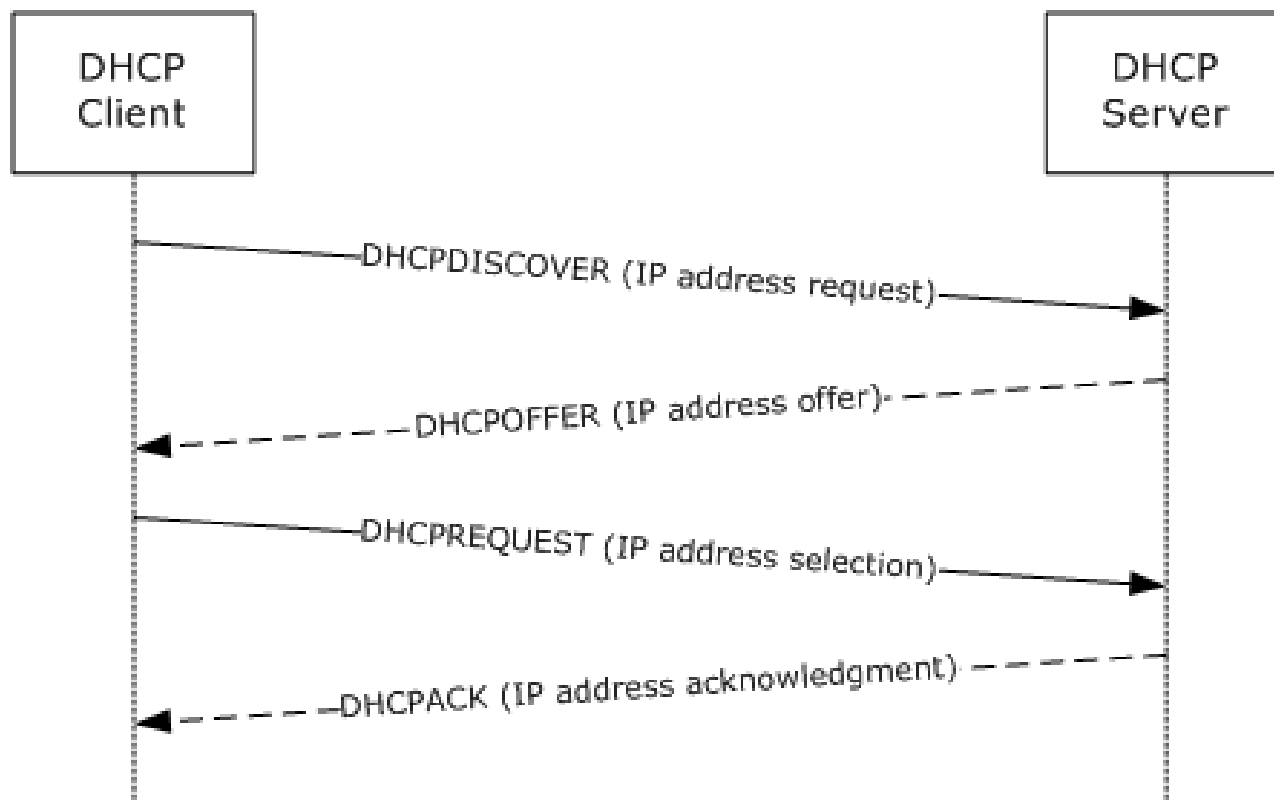
- Because DHCP has no built-in authentication, attackers can set up “rogue DHCP servers.”
- Extensions like **DHCP snooping** (on switches) and **authentication options** have been developed to mitigate this.

# DHCP – meet DORA

Not this one...



- Dynamic configuration of IP addresses (and other network configuration parameters)



# ISC KEA

ISC KEA is an open-source **Dynamic Host Configuration Protocol (DHCP) server** developed by the **Internet Systems Consortium (ISC)**. It is designed as a modern replacement for the older **ISC DHCP server** and is optimized for performance, scalability, and modularity.

Main sections:

- **interfaces:** The network interface to listen on (e.g., eth0)
- **lease-database:** Stores lease information (memfile for in-memory or mysql/postgresql for persistence)
- **subnet4:** Defines a subnet (for instance, 192.168.1.0/24)
- **pools:** Specifies the range of dynamic IP addresses (for instance, 192.168.1.100 - 192.168.1.200)
- **option-data:** Adds DHCP options like gateway (routers) and DNS servers.

# Our exercise

This is also an exploratory exercise. Not everything that you need to do is 100% laid out in the exercise description.

You'll need to:

- Look in the documentation
- Debug (`sysctl status <service>` might come in hand, as well as `journalctl -u <service>`)
- Think